To integrate accessibility in Java based products/applications after *javax.accessibility* class can be used. Microsoft products and browsers use Microsoft Active Accessibility APIs or MSAA. In GNOME, two types of APIs are defined – for the client side, it is Assistive Technology Service Provider Interface (AT-SPI)) and for the server-side, it is the Accessibility Toolkit (ATK).

There are browser extension plugins like Firevox (*http://firevox.clcworld.net/features.html*) which work in Firefox browsers; Chromevox for Chrome Browser (*http://www.chromevox.com/*) and a similarly specific screen reader for Safari, the Mac based browser. These browser based screen readers read out the content directly from the browser and separate software need not be installed.

The screen reader queries the browser accessibility APIs to fetch the information from the Document Object Model (DOM) present in the browser and conveys it to the screen reader users. Each implementation of an accessibility API defines its own roles, which are used by the browsers.

WAI-ARIA (Web Accessibility Initiative - Accessible Rich Internet Applications) is the Accessible Rich Internet Applications Suite. When it is added to the Web page, the browser detects the ARIA attribute and updates the information on its accessibility API. The screen reader queries the browser's accessibility APIs and gets the updated information. ARIA adds dynamic content and advanced technologies which have been developed using Ajax, HTML and JavaScript, and other latest technological improvements. WAI-ARIA describes many new navigation techniques to mark regions and also includes common structures like menus, primary content, secondary content, banner information, etc.

## Simple accessibility checks for websites

For screen readers to work properly, the accessibility APIs are to be implemented properly. Even for basic websites, the simple, preliminary, accessibility checks and guidelines listed below can be followed:

- All functions to be available via the keyboard
- The display setting should be visible for all contrast modes. Provide enough contrast between the text and its background
- Every image/picture should have meaningful alternate text
- For every complex user function, there should be an associate keyboard shortcut
- The page title should be meaningful and should describe the accessed Web page
- Heading levels should have a meaningful hierarchy
- When texts are zoomed or enlarged, the Web page's design or functionality should not get broken
- Labels and form control should be keyboard accessible and the labels need to be proper
- All the media in the Web page ought to be accessible using audio, video and text formats
- Columns should be added in data tables and row headers should be identified using tags
- If possible, CSS should be used for tables
- Table cells should be associated with proper IDs and headers
- Repeatedly flashing images should not be used. Never use the strobe effect
- Plug-in details should be explained properly and downloading the plugin should be accessible-friendly
- Any of the documents formats like PDFs, etc, should be accessible with assistive technologies

These guidelines or checks can easily be integrated at the time of designing a new website.

## Tools to check accessibility

There are many free and commercial tools that can be used to test the website's accessibility.

- *http://www.w3.oorg/WAI/ER/tools/complete*
- *http://www.colostate.edu/Dept/ATRC/tools.htm*
- *http://www.webaim.org/articles/tools/*

## A few thoughts on usability and accessibility

Usability and accessibility testing should be done for all electronic and information technology related products.

Even though guidelines are defined, the questions that remain are, do we have enough motivation as business owners or product designers to make websites and products globally and uniquely accessible? And do we have the required skill sets to test the accessibility of a product or website?

Ignorance is an easy excuse to avoid addressing problems but awareness is the first step forward. According to NASSCOM, the software sector reached aggregated revenues of US$ 147 billion in 2015 and with a possible 2,000 software start-ups a year by 2020, its high time the Web usability and accessibility guidelines are taken into consideration, properly.

Many of us may not be directly involved in the design phase of Web based products and our governments may be trying their best to pass some accessibility related bill or law. But we need not wait for others to act; and should instead try to contribute to the cause in any way, even if it seems insignificant at times. END

### References


[1] *https://en.wikipedia.org/wiki/Usability*
[2] *https://en.wikipedia.org/wiki/Accessibility*
[3] *http://www.w3.org/standards/webdesign/accessibility*



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